

Single-Cell Foundation Models for Cell Type Annotation and Perturbation Prediction: A Comprehensive Survey

Abstract

The advent of single-cell foundation models (scFMs) has fundamentally transformed computational biology, shifting the paradigm from task-specific architectures to generalizable, pre-trained systems capable of zero-shot inference across diverse biological contexts. This survey provides a comprehensive review of the state-of-the-art in scFMs, specifically focusing on two critical downstream applications: cell type annotation and perturbation prediction. We synthesize evidence from 300 recent academic papers to construct a detailed taxonomy of methodological families, ranging from transformer-based masked language modeling and state-space models to generative flow matching and agentic large language model (LLM) frameworks. We analyze the evolution of architectural innovations, including the integration of gene regulatory networks, multi-modal alignment with histopathology and text, and the emergence of discrete diffusion processes for robust count modeling. Furthermore, we critically evaluate the mechanisms by which these models capture biological semantics, examining mechanistic interpretability studies that reveal how attention heads and sparse autoencoders encode regulatory logic versus co-expression patterns. Finally, we discuss persistent challenges in out-of-distribution generalization, the “fluent but not faithful” gap in LLM-based reasoning, and the urgent need for standardized benchmarking frameworks to assess causal validity and clinical deployability.

1. Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled the dissection of cellular heterogeneity at unprecedented resolution, generating massive atlases comprising millions of cells across tissues and species. However, the high dimensionality, sparsity, and technical noise inherent in these datasets have historically necessitated specialized, often siloed, machine learning pipelines for tasks such as cell type annotation and the prediction of cellular responses to perturbations. The recent emergence of single-cell foundation models (scFMs) represents a paradigm shift, leveraging self-supervised pre-training on vast corpora of unlabeled data to learn universal representations of cellular states [1], [2], [3]. These models, inspired by the success of large language models (LLMs) in natural language processing, treat genes as tokens and cells as sentences, enabling the transfer of biological knowledge across domains without extensive retraining [4], [5].

The motivation for adopting foundation models in single-cell genomics is twofold. First, the sheer scale of available data—now exceeding hundreds of millions of cells—demands architectures capable of scaling laws similar to those observed in vision and language [1], [6]. Second, the biological complexity of gene regulation requires models that can internalize non-linear dependencies and contextual interactions that traditional linear methods fail to capture [7], [8]. Early approaches relied heavily on variational autoencoders (VAEs) and graph neural networks (GNNs) for dimensionality reduction and batch correction [9], [10]. However, modern scFMs increasingly utilize transformer architectures with masked language modeling (MLM) objectives, such as scGPT and Geneformer, which have demonstrated superior performance in capturing long-range gene dependencies and co-expression structures [11], [12].

This survey focuses on two pivotal applications where scFMs have shown transformative potential: cell type annotation and perturbation prediction. In cell type annotation, the goal is to assign biological identities to unlabeled cells, a task complicated by the existence of rare cell types, con-

tinuous differentiation trajectories, and cross-species variations [13], [14]. Recent advances have integrated LLMs to leverage external biological knowledge, enabling zero-shot annotation through semantic reasoning rather than purely statistical matching [15], [16]. In perturbation prediction, models aim to forecast transcriptional changes induced by genetic knockouts or drug treatments, a capability essential for drug discovery and functional genomics [17], [18]. Here, generative models based on flow matching and optimal transport have emerged as powerful tools for modeling the distributional shifts associated with cellular interventions [19], [20].

Despite these advances, significant challenges remain. Mechanistic interpretability studies suggest that while scFMs capture rich biological features, their attention mechanisms often reflect co-expression rather than direct causal regulation, raising concerns about their utility for inferring gene regulatory networks (GRNs) [21], [22]. Furthermore, benchmarking efforts reveal that simpler linear models can sometimes outperform complex foundation models on specific perturbation tasks, highlighting a gap between representation quality and task-specific utility [23], [24]. This paper synthesizes the current landscape of scFMs, categorizing methodological innovations, evaluating their efficacy through extensive empirical evidence, and outlining future directions for achieving truly universal and interpretable virtual cell models.

2. Method

The methodological landscape of single-cell foundation models is diverse, encompassing a wide array of architectural designs, training objectives, and integration strategies. We organize this section into coherent methodological families, analyzing the technical underpinnings of representation learning, generative modeling for perturbation, multi-modal integration, and the emerging role of agentic AI.

2.1. Architectural Paradigms in Single-Cell Representation Learning

The core of any scFM is its architecture, which dictates how high-dimensional, sparse gene expression vectors are encoded into latent representations. The field has evolved from standard autoencoders to sophisticated transformer and state-space models.

2.1.1. Transformer-Based Masked Language Modeling The dominant paradigm in scFMs adapts the transformer architecture, originally designed for natural language, to genomic data. These models typically employ a Masked Language Modeling (MLM) objective, where a subset of gene tokens is masked, and the model predicts their expression values or identities based on the context provided by remaining genes [1], [25].

scGPT and Geneformer: Representative models like scGPT and Geneformer utilize standard transformer encoders to process rank-encoded or binned gene expression sequences. scGPT introduces a student-teacher dual encoder architecture in variants like Cell-JEPA, where a student network predicts latent representations of masked inputs from a teacher network processing unmasked data, thereby mitigating the encoding of technical noise [26]. Geneformer employs rank-value encoding to predict masked gene indices, effectively learning the relative importance of genes within a cell’s transcriptional program [1]. Scaling analyses indicate that increasing pre-training data to over 100 million cells significantly enhances generalization to unseen donors and diseases, adhering to power-law scaling laws similar to those in NLP [1], [6].

Innovations in Tokenization and Attention: A critical challenge in applying transformers to scRNA-seq is the handling of continuous expression values and the permutation invariance of

gene sets. Methods like Exceiver adapt the Perceiver IO architecture, using cross-attention to map variable-length gene inputs to a fixed latent process, employing Discrete Noise Masking (DNM) to handle count data specifically [4]. To address the quadratic complexity of self-attention, newer models like scHyena replace attention layers with bidirectional Hyena operators, achieving $\mathcal{O}(L \log L)$ complexity while capturing long-range dependencies in full-length gene profiles [27]. Similarly, xTrimoGene employs an asymmetric encoder-decoder design where the encoder processes only non-zero tokens, reducing computational load by an order of magnitude while maintaining biological fidelity [28].

Optimization and Stability: Training stability remains a concern for deep transformers on sparse data. Recent work introduces Sigmoid Attention mechanisms, replacing softmax normalization with element-wise sigmoid functions to prevent gradient explosion and enable block-sparse computation, achieving significant speedups on modern hardware [29]. Furthermore, the choice of objective function is pivotal; the Whole-Cell Expression Decoder (WCED) objective, which forces the model to reconstruct the full cell profile from a CLS token bottleneck, has been shown to outperform standard MLM in zero-shot clustering tasks by enforcing a more unified global representation [25].

2.1.2. State Space Models and Linear Complexity Architectures To overcome the computational bottlenecks of transformers, State Space Models (SSMs) have gained traction for their linear scaling properties.

GeneMamba and scMamba: GeneMamba utilizes a bidirectional Mamba architecture with a learnable gating mechanism to integrate forward and reverse gene contexts, effectively capturing upstream and downstream regulatory influences with linear complexity [2]. Extending this to multi-omics, scMamba employs patch-based tokenization where genomic regions are treated as words, utilizing State Space Duality (SSD) to process raw features without the need for highly variable gene (HVG) selection, thus preserving positional information often lost in feature filtering [30]. These models demonstrate that SSMs can achieve competitive or superior performance to transformers in batch integration and cell type classification while being significantly more efficient [2], [30].

Hybrid Architectures: Some approaches combine the strengths of different architectures. ChromFound, designed for scATAC-seq, integrates Window Partition Self-Attention (WPSA) for local enhancer-promoter interactions with Mamba layers for global context, creating a hybrid encoder that balances local precision with global efficiency [31]. Similarly, PanFoMa decouples local Transformer encoding from global Mamba processing, using a dynamic sorting mechanism to reorder genes based on global cell state context, thereby preserving biological relevance better than static ordering [32].

2.2. Generative Modeling for Perturbation Prediction

Predicting cellular responses to perturbations (e.g., drug treatments, gene knockouts) requires modeling the transition between cellular states. This has led to the adoption of advanced generative frameworks, moving beyond simple regression to distributional modeling.

2.2.1. Flow Matching and Optimal Transport Flow matching and optimal transport (OT) provide a rigorous mathematical framework for modeling the continuous trajectories of cell state transitions.

Conditional Flow Matching: Methods like CFM-GP and PRiMeFlow model perturbations as continuous trajectories defined by neural vector fields. CFM-GP conditions the flow on cell type embeddings, allowing a single model to learn perturbation principles across diverse cellular contexts while preserving context-specific nuances [33]. PRiMeFlow operates directly in high-dimensional gene expression space using a U-Net parameterized velocity field, enabling accurate modeling of higher-order moments and variance, which is crucial for capturing the heterogeneity of perturbation responses [34].

Distributional Alignment: Recognizing that perturbations shift entire population distributions rather than just mean expression, methods like scDFM and PerturbDiff integrate Maximum Mean Discrepancy (MMD) regularization into flow matching objectives. scDFM aligns terminal generated distributions with ground-truth populations, effectively capturing complex population shifts that point-estimate models miss [18]. PerturbDiff takes this further by defining the diffusion process directly over probability distributions in a Reproducing Kernel Hilbert Space (RKHS), treating cell populations as distribution-valued random variables to capture population-level response shifts [19].

Optimal Transport Solvers: Optimal Transport offers a geometric approach to mapping control states to perturbed states. Departures approximates entropy-regularized OT via Neural Schrödinger Bridges, using minibatch OT to replace ill-posed bidirectional iterative fitting, thus enabling scalable alignment of control and perturbed distributions [20]. Similarly, Conditional Monge Gap (CMonge) learns a global parametric transport map conditioned on drug and dosage contexts, allowing a single model to handle hundreds of conditions without performance degradation [35]. For unpaired data, Unlasting utilizes Dual Diffusion Implicit Bridges (DDIB) to align unperturbed and perturbed distributions via a shared Gaussian prior, establishing implicit pairings without requiring matched control-perturb pairs [36].

2.2.2. Diffusion Models and Latent Generative Processes Diffusion models have emerged as powerful tools for generating realistic post-perturbation expression profiles, particularly when modeling the stochastic nature of biological responses.

Discrete and Continuous Diffusion: While many models operate in continuous space, the discrete nature of UMI counts motivates discrete diffusion approaches. Lingshu-Cell operates in a discrete token space via quantization, using a masked discrete diffusion process that naturally accommodates the zero-inflated sparsity of scRNA-seq data [37]. scDiVa implements continuous-time masked discrete diffusion, aligning the forward masking process with biological dropout mechanisms and utilizing entropy-normalized serialization to prioritize discriminative genes [38].

Latent Space Diffusion: To reduce computational cost, models like scPPDM and HINGE operate in frozen latent spaces. scPPDM employs a latent-space variance-preserving diffusion model guided by decomposable attention mechanisms to fuse pre-perturbation states and drug structures, decoupling state and drug conditions to prevent gradient interference [39]. HINGE adapts a frozen scFM backbone using SoftAdaLN modules to inject histology context into a masked diffusion process, enabling the generation of spatial gene expression profiles from images [40].

Causal and Counterfactual Generation: Addressing the need for causal interpretability, Latent Causal Diffusion (LCD) frames gene expression as a stationary stochastic process governed by SDEs, decoupling biological noise from technical measurement noise to infer causal dynamics from static snapshots [41]. CELLXPRT employs a training-free Metropolis-Hastings sampler to steer a multi-omics foundation model towards perturbed states on the pre-training manifold, preserving biological

variability while generating in-silico perturbations [42].

2.3. Integration of Biological Priors and Knowledge Graphs

A key differentiator of advanced scFMs is the explicit incorporation of biological priors, such as Gene Regulatory Networks (GRNs) and pathway knowledge, to guide representation learning and improve interpretability.

2.3.1. Graph-Enhanced Foundation Models Integrating GRNs directly into the model architecture allows for the propagation of regulatory signals across gene networks.

Structure-Aware Integration: GRNFormer constructs hierarchical GRNs using scATAC-seq and scRNA-seq data, processing them with GraphSAGE to mitigate degree imbalance before fusing structural and sequence features via adaptive cross-attention [11]. GenoHoption converts gene sequences into graphs and applies graph diffusion with decay constraints to expand receptive fields without parameter explosion, effectively integrating explicit gene network structures into the foundation model [43]. HEIST models tissue as spatial cellular neighborhood graphs where each cell contains an internal GRN graph, executing intra-level and cross-level message passing to capture context-specific gene embeddings modulated by local microenvironments [44].

Dynamic and Contextual GRNs: Recognizing that GRNs are dynamic, methods like Marlene employ GRU-evolved self-attention weights to capture regulatory weight changes over time, reconstructing dynamic graphs from temporal single-cell data [45]. TxPert integrates four complementary graph sources (STRINGdb, GO, PxMap, TxMap) via an Exphormer-MG architecture, demonstrating that combining curated databases with screening-derived graphs significantly enhances prediction accuracy for unseen perturbations [46].

2.3.2. Knowledge-Guided Learning and Distillation Beyond architectural integration, biological knowledge can be injected via loss functions or distillation strategies.

Privileged Information Learning: KINDLE employs a teacher-student distillation framework where a teacher model integrates temporal expression and prior masks to predict future states, distilling this regulatory logic into a student model that achieves prior-free inference [47]. InfoSEM replaces weak Laplace priors on adjacency matrices with informative priors derived from BioBERT gene embeddings, guiding unsupervised GRN inference without costly ground truth labels [48].

Semantic and Ontological Alignment: LangCell aligns scRNA-seq data with textual descriptions from OBO Foundry ontologies via a dual-encoder architecture, injecting unstructured expert knowledge into the representation space to enable true zero-shot cell identity understanding [49]. scCello incorporates cell ontology structure via a relational alignment loss using Personalized PageRank, forcing latent spaces to reflect biological lineage and improving performance on unseen cell types [50]. CellRefine introduces an intermediate post-pretraining stage utilizing marker gene prototypes and lineage-aware regularization to inject biologically grounded inductive biases before downstream fine-tuning [51].

2.4. Multi-Modal and Cross-Modal Integration

The integration of scRNA-seq with other modalities, such as histopathology images, spatial coordinates, and natural language, enables a more holistic understanding of cellular states.

2.4.1. Image-Transcriptomics Alignment Bridging the gap between morphology and gene expression is critical for spatial biology and digital pathology.

Contrastive and Generative Alignment: PAST employs a dual-encoder contrastive pretraining strategy, aligning ViT image patches with hybrid Transformer-GAT gene encoders using a similarity-adjusted contrastive loss that incorporates intra-modality cell similarity [52]. ST-Align decomposes data into spot and niche levels, utilizing specialized adaptive encoders and an attention-based fusion network to achieve three-target alignment across spot, niche, and fused feature spaces [53]. For generative tasks, GeneFlow employs rectified flow to translate single-cell gene expression to histopathological images, establishing continuous bijective mappings that avoid the stochastic trajectory errors of standard diffusion models [54].

Microenvironment Awareness: MAD (Microenvironment-Aware Distillation) utilizes a dual-view joint self-distillation framework to align morphological and microenvironmental views, capturing the transcriptomic manifold emerging from both intrinsic morphology and extrinsic context [55]. CellSymphony aggregates intermediate spatial tokens from histology models to create nucleus-centric morphology embeddings, fusing them with transcriptomic embeddings via bidirectional contrastive alignment [56].

2.4.2. Language-Cell Multimodal Models Integrating natural language allows models to leverage the vast biological knowledge encoded in LLMs.

Text-Expression Fusion: scMMGPT integrates a scGPT cell encoder with an LLaMA-2 text decoder via bidirectional Q-Formers, enabling mutual translation between expression profiles and semantic descriptions [57]. Cell2Text freezes a Geneformer encoder and projects gene-level embeddings into an LLM semantic space, training the model to generate coherent natural language descriptions of cells, which forces the learning of richer cellular representations than direct classification [58].

Open-World Knowledge Augmentation: OKR-CELL employs an LLM-based Retrieval-Augmented Generation (RAG) pipeline to enrich cell metadata with open-world biological context, implementing a Cross-modal Robust Alignment objective to dynamically weight sample reliability during pre-training [59]. scInterpreter initializes gene embeddings using GPT-3.5 processing of NCBI descriptions, projecting them into an LLM input space to leverage broad generalized knowledge for superior supervisory signals [15].

2.5. Agentic AI and LLM-Driven Reasoning

Beyond passive representation learning, there is a growing trend towards using LLMs as active agents for reasoning, hypothesis generation, and automated analysis.

2.5.1. Autonomous Agents for Annotation and Discovery Agentic frameworks orchestrate tools and reasoning loops to perform complex biological tasks.

Iterative Reasoning Loops: CellMaster executes a four-stage iterative loop involving hypothesis generation, dynamic marker selection, and multi-level evaluation, emulating expert biological reasoning workflows to perform zero-shot annotation with interpretable rationales [60]. SCPI-LOT implements “Omics-Native Reasoning” via a Problem-to-Text Converter that compresses high-dimensional matrices into semantic sketches, enabling iterative claim-evidence loops that lift annotation accuracy and reduce trajectory graph-edit distance [61].

Multi-Agent Systems: MAT-Cell decomposes annotation into a Reverse Verification Query (RVQ) for premise construction and a Syllogistic Derivation Tree (SDT) debate among verifier agents, enforcing logical consistency and grounding inputs in observed differentially expressed genes to prevent hallucination [62]. CellTypeAgent employs a two-stage hybrid verification mechanism, combining LLM reasoning ranks with database-derived expression statistics to mitigate hallucination and improve trustworthiness [63].

2.5.2. LLMs as Oracles and Reasoners LLMs are also used as “oracles” to interpret data or as reasoners to solve specific biological puzzles.

Reasoning Puzzles: Cell-o1 formulates cell annotation as a batch-level reasoning puzzle, training models via Group Relative Policy Optimization (GRPO) on distilled expert-like reasoning traces to enable emergent self-reflection and curriculum learning [64]. SUMMER discretizes continuous perturbation responses into binary QA tasks, utilizing a retrieve-summarize-answer framework to traverse structured biological knowledge graphs for interpretable answers [65].

Benchmarking and Evaluation: SC-ARENA defines a “Virtual Cell” abstraction encapsulating static attributes and dynamic methods, operationalizing evaluation via natural language tasks to reveal the “fluent but not faithful” gap where LLMs perform well on text generation but fail at ontological precision [66]. BaisBench evaluates AI scientists on cell type annotation and scientific discovery tasks, revealing that current agents substantially underperform human experts, particularly in translating high-level plans into executable code [67].

2.6. Mechanistic Interpretability and Circuit Analysis

Understanding *how* scFMs make predictions is as crucial as their performance. Recent work applies techniques from mechanistic interpretability to dissect the internal workings of these models.

2.6.1. Sparse Autoencoders and Feature Decomposition Sparse Autoencoders (SAEs) are used to decompose dense latent representations into interpretable, monosemantic features.

Feature Atlases: Application of TopK SAEs to models like Geneformer and scGPT reveals massive superposition, decomposing residual streams into overcomplete dictionaries where features organize into biologically coherent co-activation modules invisible to standard SVD [12], [68]. scFeatureLens demonstrates that SAEs can recover hidden variables learned in superposition if they directly impact observables, linking sparse features to Gene Ontology terms via automated pipelines [69].

Circuit Tracing: Exhaustive circuit mapping via causal mediation analysis exposes heavy-tailed hub architectures in scFMs, where a small fraction of features governs a large portion of the network’s activity, and redundancy increases monotonically with depth [22]. Transcoder-based circuit analysis replaces MLP layers with wide, sparsely activated transcoder layers to resolve polysemanticity, mapping internal features to specific gene regulatory programs [70].

2.6.2. Attention and Geometric Analysis Analyzing attention patterns and embedding geometry provides insights into the biological logic encoded by transformers.

Attention Mechanisms: Systematic evaluations reveal that attention patterns in scFMs encode layer-specific biological structure, with early layers capturing protein-protein interactions and late layers reflecting regulation, yet they often provide zero incremental value over trivial baselines

for GRN inference, suggesting they capture co-expression rather than unique regulatory signals [21], [71]. Spectral geometry audits via SVD identify orthogonal axes in transformer layers that encode distinct biological modalities, showing progressive compression of gene representations onto dominant axes [71].

Manifold and Geometry: The “Cell-as-Token” analogy maps transcriptomic profiles to high-dimensional vector spaces, identifying low-dimensional manifolds and biological polysemy analogous to semantic structures in language [72]. Geometric coherence analysis of CRISPR perturbations reveals that pleiotropic regulators incur a “geometric tax,” producing large but incoherent shifts, while lineage-specific factors drive coherent trajectories [73].

2.7. Benchmarking and Evaluation Frameworks

Robust evaluation is critical for assessing the true capabilities of scFMs. The field has moved towards standardized benchmarks that test generalization, robustness, and biological fidelity.

2.7.1. Perturbation and Drug Response Benchmarks Hierarchical and Distributional

Metrics: PerturBench introduces rank-based metrics to detect mode collapse and assess perturbation specificity, arguing that traditional fit metrics like RMSE fail to distinguish non-specific predictions [74]. VCBench establishes embedding-defined unseen-cell splits to reveal significant performance drops compared to random splits, exposing limited robustness to cellular context shifts [75].

Drug Response Prediction: scDrugMap integrates ten foundation models into a unified pipeline, demonstrating that pooled-data evaluation yields superior performance compared to cross-data settings, indicating a reliance on broad data exposure rather than true zero-shot generalization [76]. scGPT-DeepCDR replaces traditional omics encoders with frozen scGPT embeddings, outperforming baselines in drug response prediction by capturing complex gene interactions more effectively [77].

2.7.2. Annotation and Clustering Benchmarks Zero-Shot and Cross-Species Evaluation:

SOAR formulates cell annotation as a question-answering task, benchmarking instruction-tuned LLMs on single-cell genomics and showing that Chain-of-Thought prompting significantly improves performance over zero-shot baselines [13]. scCluBench evaluates 14 algorithms across 36 datasets, finding that traditional methods fail on high-sparsity large-scale data while graph-based models suffer from over-smoothing unless utilizing cut-informed mechanisms [78].

Standardized Resources: scUnified provides a standardized resource with uniform preprocessing and annotation, enabling direct application of diverse models and revealing that deep learning methods achieve perfect classification on certain benchmarks when data is properly curated [79]. Parameter-free representations like scTOP, based on careful z-score normalization and linear projection, have been shown to outperform complex foundation models on downstream benchmarks, suggesting that much of the biological signal resides in low-complexity linear subspaces [24].

3. Challenges and Outlook

Despite the rapid progress in single-cell foundation models, several critical challenges hinder their widespread adoption and reliable deployment in clinical and research settings.

3.1. The Generalization Gap and Out-of-Distribution Robustness

A recurring theme in recent benchmarks is the fragility of scFMs when faced with out-of-distribution (OOD) data. While models excel in pooled-data settings where training and test distributions overlap, their performance degrades significantly in cross-data or unseen-cell-line scenarios [76], [75]. For instance, embedding-based splits in perturbation benchmarks reveal that models often fail to generalize to cellular contexts not seen during training, suggesting they memorize dataset-specific biases rather than learning universal biological principles [75]. This limitation is particularly acute in drug response prediction, where models struggle to predict effects for compounds outside the training distribution or for cell lines with distinct regulatory architectures [17], [46]. Future research must prioritize the development of invariant representation learning techniques that disentangle biological signal from technical and contextual confounders to achieve true zero-shot generalization.

3.2. Interpretability vs. Performance Trade-offs

While scFMs achieve state-of-the-art performance, their “black-box” nature remains a significant barrier to biological insight. Mechanistic interpretability studies indicate that attention mechanisms often capture co-expression patterns rather than direct causal regulatory relationships, limiting their utility for inferring Gene Regulatory Networks (GRNs) [21], [22]. Although methods like Sparse Autoencoders and circuit tracing offer glimpses into model internals, a large fraction of extracted features remain uninterpretable due to gaps in current biological knowledge [68], [22]. Furthermore, the “fluent but not faithful” behavior of LLM-based agents, where models generate plausible but factually incorrect biological rationales, poses risks for automated discovery [66], [67]. Bridging this gap requires the integration of stronger causal inductive biases and the development of evaluation metrics that prioritize biological fidelity over mere predictive accuracy.

3.3. Computational Efficiency and Scalability

The computational cost of training and inference for large-scale scFMs is prohibitive for many research groups. Transformer-based models suffer from quadratic complexity with respect to sequence length, although innovations like State Space Models (e.g., Mamba) and sparse attention mechanisms are beginning to alleviate this burden [2], [27], [29]. However, multi-modal models that integrate high-resolution images or long-context genomic sequences still demand massive GPU resources, limiting accessibility [31], [80]. Additionally, the inference latency of iterative generative models, such as diffusion processes and agentic reasoning loops, hinders their application in real-time clinical settings [38], [60]. Developing lightweight, efficient architectures and distillation techniques will be essential for democratizing access to these powerful tools.

3.4. Data Heterogeneity and Standardization

The quality and consistency of training data remain fundamental bottlenecks. Single-cell datasets exhibit vast heterogeneity in protocols, sequencing depths, and annotation standards, leading to models that are sensitive to these variations [79], [81]. While methods like scUnified and HR-VILAGE-3K3M attempt to standardize preprocessing and metadata, the lack of universal gold-standard benchmarks and ground-truth GRNs complicates model validation [79], [82], [83]. The reliance on incomplete or biased knowledge bases for GRN inference and LLM augmentation further propagates errors and limits the discovery of novel biology [11], [84]. Community-wide efforts to curate high-quality, multi-modal atlases with standardized annotations and experimental ground truths are urgently needed.

3.5. Safety, Ethics, and Clinical Deployment

As scFMs move towards clinical applications, such as drug response prediction and patient stratification, issues of safety and ethics become paramount. The potential for hallucination in LLM-driven agents and the lack of uncertainty quantification in many generative models pose risks for clinical decision-making [63], [85]. Models must be equipped with robust uncertainty estimation mechanisms, such as evidential learning or conformal prediction, to flag unreliable predictions [86], [87]. Furthermore, the use of patient data for training raises privacy concerns, necessitating the development of federated learning frameworks and differential privacy techniques tailored for single-cell genomics [88]. Ensuring that these models are fair, unbiased, and transparent is critical for their ethical deployment in healthcare.

4. Conclusion

Single-cell foundation models have ushered in a new era of computational biology, offering unprecedented capabilities for cell type annotation and perturbation prediction. By leveraging massive-scale pre-training, advanced generative frameworks, and multi-modal integration, these models are beginning to capture the complex dynamics of cellular systems. However, the path to truly universal and interpretable virtual cells is fraught with challenges. Addressing the generalization gap, enhancing mechanistic interpretability, improving computational efficiency, and establishing rigorous benchmarking standards are critical priorities for the field. As the community converges on standardized datasets and evaluation protocols, and as architectural innovations continue to mature, scFMs hold the promise of transforming our understanding of cellular biology and accelerating the discovery of novel therapeutics. The future lies not just in larger models, but in smarter, more causal, and more trustworthy systems that can reason about biology with the depth and nuance of human experts.

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